

Acquisition Is Not Closure: Fail-Closed Control for Open-World Scientific-Literature Discovery

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ABSTRACT

Scientific-literature discovery combines evidence acquisition with decisions about when a search may stop. These functions are often evaluated as one policy, although a locally exhausted or unavailable route does not establish that the scientific task is complete. We introduce a fail-closed control layer that separates route stopping from task closure, screens route distinctness by provenance, distinguishes content identity from question-conditioned processing, and retains unresolved material routes as open obligations. A 390-task complete-gold index supplies descriptive mechanism evidence only: mean recall is 0.979487 with no premature task closure, and the study does not establish superiority. Figure 3 traces the controlled index's recall-resource frontier. In a 400-case exact-contract battery, the controller matches 400 of 400 declared closure labels; an authored comparator assembled from prior mechanisms, excluding unavailable material routes from the closure denominator, matches 250 of 400. An information-equivalent comparator also matches 400, bounding the claim to the closure contract. A descriptive TREC-COVID comparison is adverse on recall and cost. Against BM25, recall@100 is lower by 0.01769, with a 95% bootstrap interval of $[-0.02729, -0.00906]$, and reads increase by 175.7%. nDCG@10 is higher by 0.1488; that secondary result cannot replace the recall and cost criteria. Density-normalized and query-conditional BEIR successors leave all five ArguAna decisions unchanged. These transformations are decision-inert under the tested rule. The contribution is therefore a bounded control method, not evidence of external retrieval superiority or open-world completeness.

Keywords: scientific literature discovery; information retrieval; systematic review; stopping rules; coverage; open-world search; fail-closed control.

1. Introduction

Scientific-literature discovery systems increasingly combine lexical and semantic retrieval, query reformulation, citation expansion, selective reading, screening, and learned stopping. Better acquisition can recover more useful material, but it cannot answer a different question by itself: what does the observed search state authorize a system to claim about completeness? A route may be locally exhausted while another route remains untried. A provider may be unavailable rather than empty. Two differently named routes may share the same backend and query derivation. A document may have been encountered before but not processed for the current scientific question. Treating these cases as ordinary negative evidence creates false closure.

This paper studies that control boundary rather than proposing a new general retriever. The method separates an acquisition policy, which selects the next search or reading action, from a closure authority, which decides whether the current evidence state permits route stopping or task closure. Four design elements implement the separation. First, structural route distinctness is screened by backend, query-derivation, and capture provenance rather than route labels. Second, route-local stopping never certifies task-global completeness. Third, unavailable, censored, or provider-invalid material routes remain explicit open obligations. Fourth, content identity is recorded separately from question-conditioned read state, preventing duplicate discovery from inflating coverage without suppressing legitimate rereading.

The empirical question is deliberately narrow: do these control elements behave in their intended directions under complete denominators and protected exact contracts, and do they support an external superiority claim? We use a complete-gold controlled index to expose premature closure and mechanism failures, then an exact-contract battery to isolate unresolved-route authority. The external stress test is

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a descriptive TREC-COVID comparison. Recall and cost both move against the proposed superiority interpretation. We retain the favorable nDCG result as an opposing observation rather than treating it as an overall superiority result.

The resulting contribution is a methods and critical system-design claim. It does not claim that finite open-world search proves completeness, that the controller outperforms established retrieval systems, or that existing retrieval, screening, stopping, and memory components are novel. Instead, it provides a composable representation of what acquisition outcomes are allowed to certify, together with controlled evidence and an adverse external test that fix the present boundary.

2. Acquisition is not closure

The central design move in this paper is to separate two problems that are often collapsed into one policy. An *acquisition policy* decides what to search, retrieve, inspect, expand or reread next. A *closure authority* decides whether the evidence state is allowed to support route-local stopping or scientific task completion. Better acquisition can improve recall and reduce cost, but it does not automatically acquire authority to declare completeness. This separation makes strong external retrieval and stopping mechanisms composable rather than adversarial to the contribution of this paper.

Let x_t denote the observable search state at step t . We write an acquisition mechanism as

$$a_t \sim A(x_t),$$

where a_t may be a lexical query, a reasoning-aware query, a fielded Boolean filter, a graph-expansion action, a selective fetch, a reranking operation, or a request for another retrieval round. No novelty is claimed here for the internal form of A . Instead, acquired evidence updates both an evidence state E_t and a typed obligation state O_t . Closure is a distinct map

$$C(E_t, O_t) \in \{\text{CONTINUE, ROUTE-STOP, OPEN, CANNOT-CHECK, TASK-CLOSE}\}.$$

A utility, relevance, confidence, sufficiency or coverage signal may influence A and may recommend a local stop, but it cannot by itself delete an unresolved obligation in O_t . This interface is where the residual contribution of this paper sits.

2.1. Absorbing strong acquisition mechanisms

The acquisition layer intentionally composes with prior mechanisms. SAGE shows that keyword-oriented agent queries can make strong lexical retrieval difficult to beat and that corpus-side metadata/keyword augmentation can further improve retrieval [10]. AgentIR shows that an agent’s reasoning trace can be used jointly with its query as a retrieval signal [3]. SIEVE shows that fielded Boolean admission, structure-rich inspection and selective section fetching can improve the accuracy–context tradeoff [24]. Breadth-first bibliography expansion can recover literature missed by API-only retrieval, while also demonstrating why citation-list recall cannot be the only evaluation axis [21]. Lexical, metadata-augmented, reasoning-aware, field-aware and citation-expansion retrieval are therefore treated as candidate strengths to compose, not as inventions of this work.

The same rule applies to continuation and stopping signals. DeepControl uses marginal information utility to regulate continued retrieval [27]; decision-theoretic policies optimize screening stops against explicit costs and payoffs [8]; HALT uses claim–evidence coverage to verify whether expected evidence has been obtained [19]; MiCP applies conformal prediction to adaptive multi-turn reasoning to maintain a target coverage guarantee while permitting early stopping [30]; and structured Search-R1 stopping judgments explicitly encode sufficiency and remaining gaps [14]. These signals and guarantees can all be useful inputs to acquisition or route-local stopping. The question asked here is narrower: what are those signals permitted to certify when routes are dependent, unavailable, censored or only locally exhausted?

2.2. The authority residual

The implementation composes four control elements: provenance-based route distinctness, separation of route stopping from task closure, explicit open states for unavailable or censored routes, and separation

Donor-composable acquisition: stronger acquisition mechanisms may improve evidence, but local signals do not confer scientific-closure authority.

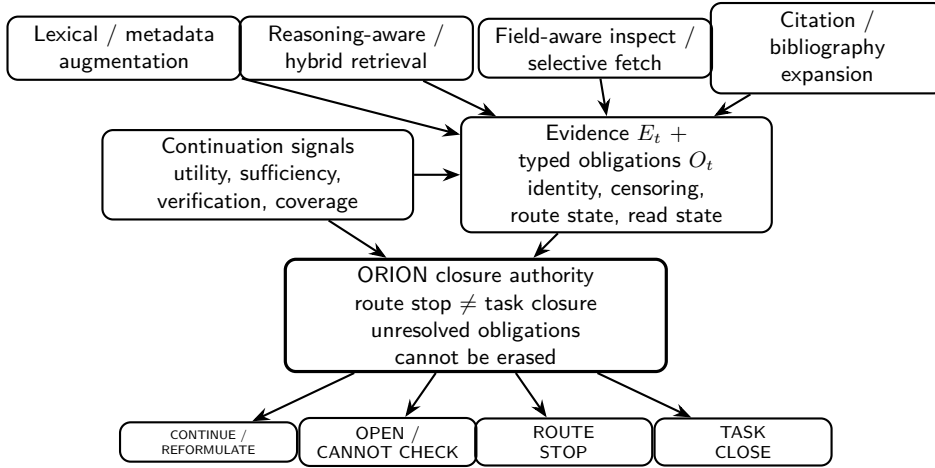


Figure 1: Acquisition mechanisms are intentionally composable: lexical/metadata augmentation, reasoning-aware or hybrid retrieval, field-aware selective fetching, citation expansion and continuation signals may all improve what is acquired. Their outputs update evidence and typed obligations, but a separate authority layer controls route stop, open and undetermined states and task closure. The figure therefore locates the residual at the meaning of acquisition outcomes rather than at a claim to have invented the strongest retriever.

of content identity from question-conditioned processing state. Prior work owns strong retrieval, coverage, sufficiency, stopping, memory, and obligation mechanisms. The narrower residual tested here is their task-level aggregation: a prospectively material route that is unavailable, censored, or provider-invalid must not disappear from the closure denominator. The exact-contract comparison changes this rule while granting the comparator the other prior mechanisms.

The resulting design is deliberately asymmetric. It is permissive about *how* evidence is acquired: a stronger retriever, query generator, expansion rule or verification signal can replace a weaker one. It is conservative about *what acquisition outcomes are allowed to mean*: local flatness, high estimated coverage, high confidence, apparent sufficiency or a provider failure cannot silently become proof of global completeness. This asymmetry is what permits the paper to absorb progress in retrieval without making its claimed authority boundary obsolete whenever a stronger retriever is published.

2.3. Empirical consequence

The decomposition also fixes the widening experiment. Candidate-generation methods should first be compared and composed under matched query/candidate budgets. Only after candidate recall is high enough to make stopping observable should the closure layer be challenged with direct ablations and strong stopping baselines. A positive external result must therefore show both acquisition quality and the incremental effect of typed closure authority; a cheaper stop that hides reachable relevant evidence is not a positive result. Conversely, a stronger substituted retriever that improves acquisition while leaving the closure constraints intact strengthens the system rather than weakening the claim.

3. Formal mechanics

This section fixes the objects the evaluation measures. Every definition below is implemented in the system under test rather than assumed: work identity and read state, structural route distinctness and coverage, and route control each correspond to a module of the implementation, named in Section 8. Where a definition is deliberately conservative we state the failure it is conservative against, because the direction of the bias is what makes the refusal a safety property rather than a preference.

3.1. Work identity and content identity

Let R be the set of raw identifiers a route can emit (DOIs, arXiv ids, OpenAlex ids, local ids, URLs). Canonicalization is a map $\varphi : R \rightarrow (\mathcal{S} \times V) \cup \{\text{UNKNOWN}\}$ sending a raw identifier to a scheme-qualified key and a rendition version, so that $\varphi(\text{arXiv:2301.00001v1})$ and $\varphi(\text{arXiv:2301.00001v2})$ share a key and differ only in V : a version is a rendition of one work, not a second work. A declared scheme is treated as a claim rather than a credential; an identifier that does not match the shape of the scheme it announces resolves to UNKNOWN instead of entering the graph as a key that can never be fetched.

A *work* is the alias-closure of canonical keys. Given identity records with alias sets $A_1, \dots, A_n$, works are the connected components of the graph on $\bigcup_i A_i$ that joins two keys sharing a record. Closure is computed as a projection over an append-only ledger rather than at write time, because an alias learned late can join two works recorded early; folding at write time would freeze the wrong partition.

Separately from work identity, each retrieved rendition carries a content digest c . Route overlap, deduplication and coverage are computed over content digests, never over retrieval identifiers: a source that re-mints identifiers would otherwise make two routes appear disjoint when they retrieved the same paper, which inflates every coverage quantity built on their overlap.

3.2. Question-conditioned read state

A read receipt is a tuple (w, c, σ, f) : work, content digest, extraction schema version, and frame f — the scientific question the extraction was performed for. Given the receipt set $\mathcal{R}$, the read decision for a candidate (w, c, σ, f) is

$$\text{read}(w, c, \sigma, f) = \begin{cases} \text{UNSEEN} & \nexists r \in \mathcal{R} : w_r = w \\ \text{CONTENT_CHANGED} & \exists r : w_r = w, \nexists r : (w_r, c_r) = (w, c) \\ \text{NEW_SCHEMA} & \exists r : (w_r, c_r) = (w, c), \nexists r : (w_r, c_r, \sigma_r) = (w, c, \sigma) \\ \text{NEW_FRAME} & \exists r : (w_r, c_r, \sigma_r) = (w, c, \sigma), \nexists r : r = (w, c, \sigma, f) \\ \text{ALREADY_READ} & \exists r \in \mathcal{R} : r = (w, c, \sigma, f). \end{cases}$$

ALREADY_READ therefore requires agreement on all four coordinates. This yields the distinction the evaluation depends on: a repeat extraction with all four coordinates equal is *duplicate processing* and is waste, while a repeat extraction under a changed content digest, schema or frame is a *legitimate reread* and is required work. Collapsing the two — treating “this paper has been seen” as “this paper has been read for this question” — is how a system reports coverage of material it only skimmed for a different purpose.

3.3. Structural route distinctness

A route capture is a tuple $\rho = (k, b, d, C)$: route kind, backend identity, query-derivation process, and the set C of captured content digests. For two captures ρ_i, ρ_j the structural-distinctness predicate is

$$\text{distinct}(\rho_i, \rho_j) \iff k_i \neq k_j \wedge b_i \neq b_j \wedge d_i \neq d_j,$$

and the verdict otherwise names the shared cause (SAME_ROUTE_KIND, SHARED_BACKEND, SHARED_QUERY_DERIVATION). Independence is thus *earned* from the generation and capture process; a distinct route label carries no evidential weight. This predicate is an eligibility screen for a coverage diagnostic. It does not establish statistical independence between the returned rankings or captures.

Proposition 1 (Direction of the conservatism). *Positive dependence between capture occasions can bias unseen-mass estimators downward. Admitting a pair that shares known generation structure can therefore produce false confidence that search is near exhaustion, whereas refusing a structurally distinct pair only forfeits a diagnostic. The screen is accordingly biased toward refusal.*

3.4. Coverage diagnostics are never authority

For a structurally distinct pair whose captures additionally satisfy the sampling assumptions required by the estimator, with sizes n_i, n_j and overlap m_{ij} , the two-source estimate is $\hat{N} = n_i n_j / m_{ij}$, undefined at $m_{ij} = 0$. Zero overlap places no upper bound on the population, so the system returns a refusal with reason rather than a large number that would read as a measurement. When several independent pairs yield

estimates, the reported value is $\max_{ij} \hat{N}_{ij}$: the largest population implies the most still unfound, which is the direction that fails safe. Every coverage estimate is marked diagnostic-only, and adaptive search is explicitly not a sequence of independent capture occasions — following up on what was just found converts singletons into doubletons and deflates the estimator toward “nearly done”.

3.5. Route control state machine

Let a route hold attempts $a_1, \dots, a_t$, each with retrieved digests D_i , novel digests $N_i \subseteq D_i$, cost, failure signatures and a backend-exhaustion flag. For a nonempty return, marginal novelty is $|N_t|/|D_t|$; a successful empty return has novelty zero. The flat streak z is the number of trailing attempts that succeeded, reported no exhaustion and returned no novel digest. Under a frozen policy with thresholds $\theta_{\text{ref}} \leq \theta_{\text{sw}}$, the decision is taken in strict priority order:

$$\delta = \begin{cases} \text{STOP (budget)} & \text{remaining budget} \leq 0 \\ \text{STOP (exhausted)} & a_t \text{ reports backend exhaustion} \\ \text{SUSPEND} & a_t \text{ carries execution-failure signatures} \\ \text{SWITCH} & z \geq \theta_{\text{sw}} \wedge \text{alternatives exist} \\ \text{REFORMULATE} & z \geq \theta_{\text{ref}} \\ \text{CONTINUE} & \text{otherwise.} \end{cases}$$

Two separations in this table are load-bearing. First, backend exhaustion and transport failure are distinct states: the former is a statement about the index, the latter about the channel, and a system that merges them converts an outage into evidence of absence. The latter yields SUSPEND, a non-terminal state carrying an open obligation, not STOP. Second, each terminal decision carries typed reopen triggers (budget increase, backend recovery or refresh, material frame change, material finding on an alternative route); a route reopens exactly when an observed trigger intersects that set.

Proposition 2 (Route-stop non-certification). *No individual route decision certifies task saturation. In the implementation every route decision, STOP included, has only route-local authority and carries an explicit declaration that it does not certify task saturation. Consequently, a route stop cannot be used by itself as a task-completeness certificate. A composition of route outcomes may support task closure only through a separately specified ensemble predicate whose route universe, material-obligation registry, provider/observability/custody guards, resource state and reopen triggers were fixed before the outcome.*

Proof. The route decision alphabet contains no task-closure authority, so projection of any one route output cannot emit a valid task certificate. The controller’s task decision is instead produced by the separately typed closure map over the full ensemble state. If that map’s registered completeness and validity guards hold, a composition can certify closure; if they do not, composition of route-local stops alone supplies no missing authority. $\square$

Task closure is therefore a separate judgement over the route ensemble. It requires that every route in the prospectively registered route universe is in a terminal or explicitly reopenable state, that no suspended or unavailable route leaves an unmet obligation, and — by Proposition 2 — that no coverage diagnostic has been used as its authority. It also requires the registered provider, observability, custody and resource-validity guards to pass as one conjunction. An unavailable route remains an open coverage obligation and is reported as such; it is never converted into evidence that nothing further exists.

3.6. What the evaluation must therefore measure

The definitions above fix four measurable error families, each with a ground-truth condition that only a complete-gold world supplies: route-stop false positives (a route stopped while it could still reach unfound relevant items), route-stop false negatives (a route continued past its exhaustion point), premature task closure (task closed while a gold item was still reachable within remaining budget), and duplicate processing versus legitimate reread as separated by the four read coordinates. Route overlap, per-route marginal contribution and structural-distinctness status are computed over content digests. Metrics whose denominator is the gold set are reported only where that denominator is complete; the conditions are stated per metric in the frozen measurement plan.

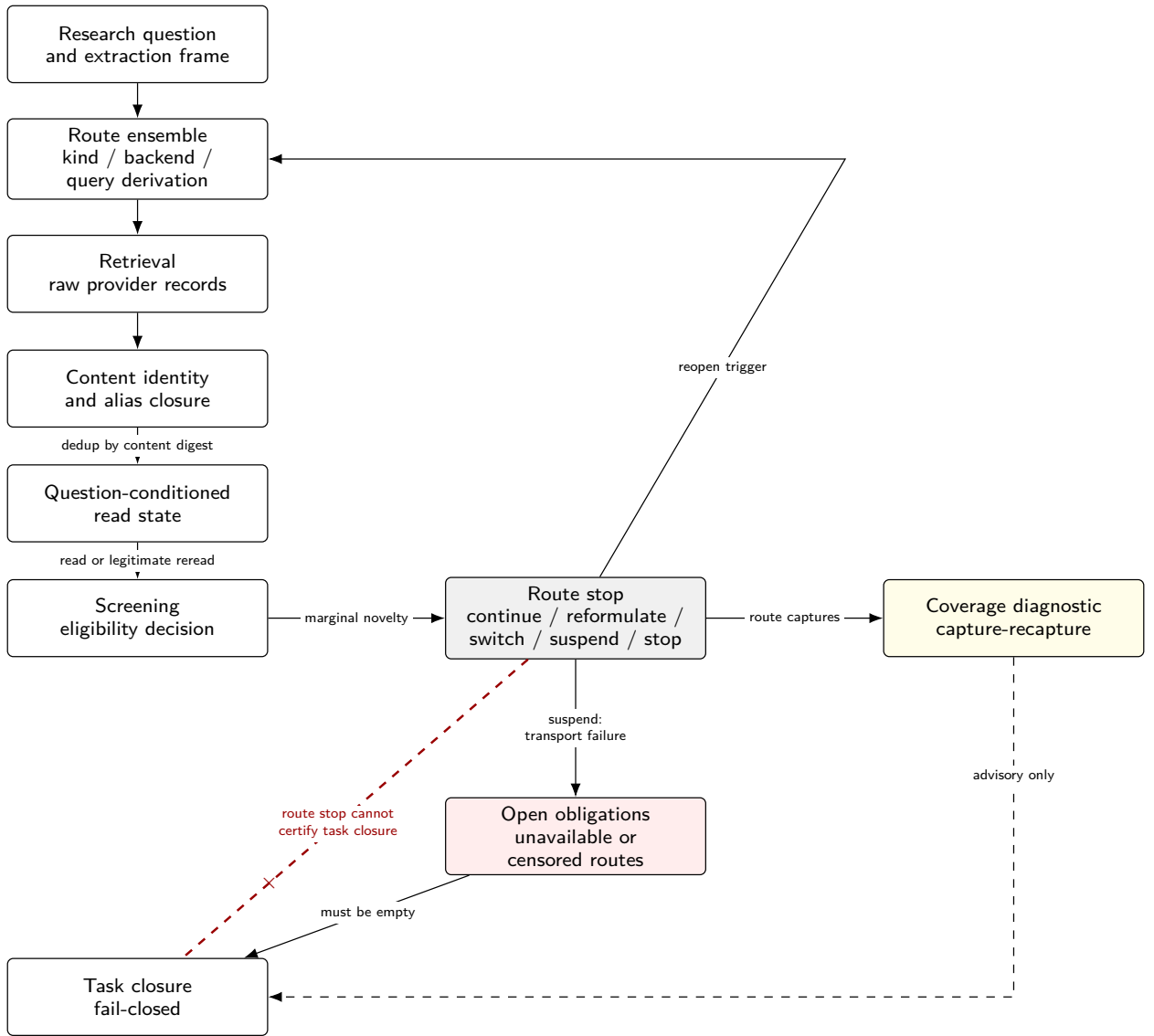


Figure 2: The discovery pipeline separates retrieval, screening, route stopping, and task closure. Solid arrows carry state updates. The dashed coverage diagnostic informs search but does not certify closure, and the crossed arrow is the forbidden promotion from a route stop to task completeness. Transport failure leaves an open obligation rather than evidence of absence.

4. Methods

This section reports the designs that generated the controlled-index, exact-contract, TREC-COVID, and BEIR evidence. Post-outcome manuscript edits do not alter their systems, tasks, endpoints, margins, budgets, exclusions, or decision rules. The versioned design, execution, and result records govern when their status differs from historical prose. Section 8 describes the archived evidence and its access conditions.

4.1. Three-valued outcomes

Every decision this paper measures is three-valued rather than binary. A task may be determined complete, determined incomplete, or left *undetermined*; a route may be determined exhausted, determined to have more to give, or left undetermined. “We could not determine this” is recorded, reported and

counted separately from “we determined this is false”, and the two are never merged. The distinction is the point: a provider outage, a missing credential or a censored route produces an undetermined outcome, and an undetermined outcome is not evidence of absence. Throughout the paper an outcome described as *undetermined* is this third value, carried in the evidence record and in every downstream count, never silently resolved to a negative.

4.2. Prospective discipline

The prospective protocol `P2.open-world-discovery.v1` is frozen and versioned before execution. A result-bearing execution is admissible only after subject code, data, evaluator, system identities, budgets and repeat seeds are bound. For the completed offline companion, the run manifest records that no outcome was accessed before the freeze, together with the exact subject revision, content hashes, 14 system identifiers, three repeat seeds and host-owned resource limits. The bound subject revision passed the full test suite before the first aggregate outcome snapshot was recorded. Publication summaries and manuscript prose may be generated after outcomes, but they may only project the frozen records; they cannot change the scientific decision rules.

4.3. Evaluation environments

Three environments are distinguished because no single one supplies both a complete denominator and open-world realism. The first supplies the only full system comparison with a complete denominator; later bounded external probes have narrower acquisition or screening authority and are reported separately.

Offline complete-gold controlled index. Recall is only measurable where the relevant set can be enumerated. The controlled index is therefore fully synthetic and self-generated. Each topic has a materialized gold set under the frozen generator, and tests recompute the world and tasks from the recorded seed and compare all committed hashes rather than trusting a hand-authored denominator. Discoverability is heterogeneous: relevant items are partitioned across lexical, semantic, citation, reformulation and restricted/provider routes; two route labels deliberately share a backend; duplicate locators share content identity; revisions share identity but change digest; extraction-question-shift cases require legitimate rereads; and unavailable-route cases distinguish transport censoring from absence. Budgets are below exhaustive probing, so route allocation and stopping remain observable variables.

The completed suite has 390 tasks over five case families, drawn from 78 topics over 1,210 documents, frozen at suite fingerprint `2f6936ba52fb`. Every candidate uses the same host-owned session. The session alone can access the world, meters queries and reads, records route stops and transport states, and evaluates claims after the candidate returns. The candidate receives only a public task view and label-free retrieved records; protected gold and the relevance rule are never passed through the system interface.

The five task variants generated from each topic share the same synthetic world and question. We therefore treat the topic, not the variant, as the independent cluster. The 390 variant-level outcomes are descriptive mechanism diagnostics; deterministic repeats test execution stability and do not increase the number of independent units.

Exact-contract battery. The exact-contract battery isolates the task-closure rule from retrieval quality. It spans five acquisition families: scientific literature, web and API evidence, dataset registries, code and artifact repositories, and experimental or measurement routes. Each family instantiates eight hostile archetypes, including an available route that remains open; an unavailable, censored, or provider-invalid material route; duplicate coverage; acquired but not question-processed content; and low-utility routes that remain material. Route order is seed-permuted, and held-out variants add unavailable non-material decoys so that a blanket refusal to close cannot pass.

The primary comparator, B1, is an authored composite of prior mechanisms rather than a published product. It receives route identity and materiality, evidence coverage, decision sufficiency, utility stopping, finite route certificates, deduplication, route state, and question-conditioned processing state. It makes the same route-local decisions as the governed controller. The sole systematic difference is task-level aggregation: B1 excludes unavailable, censored, or provider-invalid material routes from the closure denominator once all available material routes have stopped. B2 allows a global sufficiency signal to authorize closure. B3 receives

the same unresolved-route semantics as the governed controller and tests whether the rule is portable to an information-equivalent product. The protected unit is the authored case. Exact scientific closure decision is agreement with the declared case label.

Descriptive TREC-COVID comparison. The external comparison uses the 50 official Round 5 TREC-COVID topics and judgments [23]. The governed multi-route arm and three comparators share the frozen corpus, topics, maximum of 300 distinct reads, six route calls, and 400 tool calls. Citation and restricted-provider routes are unavailable in this collection and remain recorded rather than being treated as empty successful routes. No separate pre-outcome record was found that binds the comparator-selection rule, the -0.02 non-inferiority margin, the 25% read reduction criterion, or the bootstrap procedure. We therefore treat the completed four-arm comparison as post hoc and descriptive. The result artifact selects the strongest observed comparator, BM25, and applies those recall, cost, and safety reference criteria. Paired topic differences are summarized with a 10,000-resample percentile bootstrap interval. nDCG@10 is reported alongside them but does not establish an overall superiority claim.

BEIR mechanism diagnostics. Three prospectively separated studies use SciFact, NFCorpus, and ArguAna with fixed corpus hashes, held-out query halves, and depths 10, 20, 50, 100, and 200. Five retrieval functions over each local corpus create route heterogeneity. The original route-aware rule compares an estimated unread-route contribution with an absolute threshold. The first successor changes only the threshold scale through development-split relevance-density normalization. The second restores the original threshold and changes only the estimator by scaling its global marginal with each held-out query’s unseen-route fraction. Its falsifier was predeclared: if all five ArguAna decisions matched the original rule, the query-conditional estimator would be recorded as decision-inert and no fourth estimator would be introduced under the same study identity.

Other external benchmark tasks. AutoResearchBench provides Wide and Deep scientific-discovery tasks [26]. Its completed probes remain bounded acquisition or transport diagnostics, not matched evidence for the full controller.

SAGE is retained as neighbouring work and as a lexical-baseline warning [10], but is not executed as if its published setting were available: the paper-linked repository is data-only at the audited revision and provides neither an official runner nor a licence. Any substitute corpus, runner or weighting rule would be a declared deviation rather than the SAGE system. MetaSyn’s separation of retrieval from screening is adopted as a measurement constraint [25]; its credential-free BM25 plus public-protocol screening probe subsequently completed on all 86 released test reviews and is reported as a bounded retrieval/screening result, not as the complete proposed controller. AgentSLR remains a protocol-driven systematic-review neighbor/baseline family rather than an official evaluator dependency [16].

Live-provider campaigns. The OpenAIRE and Crossref campaigns freeze gold-blind candidate pools and retain request, query-derivation, timestamp, raw-response, and typed transport records. Their provider-validity gates failed, so their numerical ties do not support a null, equivalence, or superiority conclusion. A separate fresh 48-task campaign was not accessed and contributes no outcome.

4.4. Systems under test

The implementation evaluated here is the ORION discovery layer; it is the artifact under test, and the rest of the paper calls it the *governed system* so that the claims read as claims about the mechanism rather than about a product. The frozen offline companion compares the governed system with six confirmatory comparators: a BM25 keyword retriever, a dense retriever, a sparse–dense hybrid, a one-pass retrieval-augmented pipeline, a single-route agentic searcher, and a protocol-driven systematic-review comparator. A seventh comparator, an adaptive multi-route exploratory searcher, was kept explicitly exploratory and was not retrofitted into the frozen confirmatory baseline set. The registry that binds each of these to its frozen system identifier is archived with the campaign.

These controlled-index systems are mechanism comparators, not claims to replicate a production BM25 engine, a particular embedding model or an external agent implementation. They consume the public synthetic route interface under identical budgets so that the variable under test is route/read/stopping

governance. Their relevance screen sees only the public question and retrieved bibliographic text. The authored synthetic text makes screening deliberately easy; the experiment is not presented as evidence about classifier quality.

Six frozen ablations remove exactly one governance mechanism or authorize one forbidden transition: route-independence checking, the question coordinate in the read ledger, route/task stop separation, unavailable-route open state, coverage-diagnostic non-authority, and content-identity deduplication. All use the same session and budgets as the governed system.

4.5. Identity, reads and stopping

A read is keyed by work identity, content digest, extraction-schema version, and scientific question. A new question or schema can therefore reopen unchanged content for a legitimate reread, while mirrors with the same work and digest do not become new evidence. The corresponding ablations remove the question coordinate or replace content identity with document locator to expose suppression or duplicate work.

Route stop and task closure are separate actions. A route may stop after its public probe space is exhausted or after a typed transport outcome, but task closure is invalid if relevant material remains reachable within budget. An unavailable route stays censored and open rather than being converted to an empty result. Coverage diagnostics may inform a route policy but have no authority to certify global completeness.

4.6. Measurements

The offline evaluator reports denominator-valid complete-gold recall, precision, missed-gold count, route use, route/content overlap diagnostics, marginal relevant gain after the first route, premature task closure, censoring, duplicate processing and legitimate rereads, plus host-owned resource counters. Claims are normalized by content identity rather than locator so mirrors cannot inflate recall. Rich per-run artifacts retain route calls, reads, stop events, authority flags and raw-artifact hashes; the publication summary binds the full 16,380-record and rich-artifact sets by SHA-256.

The external measurement plan additionally defines official Wide/Deep scores, screening metrics and ranking-only metrics where their denominators and task semantics support them. Those definitions do not imply that an external result exists.

4.7. Statistics and authority

The controlled-index outcomes are reported descriptively because the five within-topic variants are dependent and the study does not establish a population-sampling model. Deterministic repeats nest within topic, task, and system and test only stable execution. The 400 exact-contract cases are also a fixed authored battery. Their stratified bootstrap interval and McNemar count describe that battery and are not population uncertainty. TREC-COVID uses the topic as the paired unit under its archived external gate. The BEIR studies use held-out queries for their fixed-corpus mechanism decisions. These authorities are not pooled or averaged.

4.8. Failures, exclusions and contamination

Candidate failures, malformed outputs, budget exhaustion, abstention and typed transport failures remain outcomes. Provider unavailability cannot become evidence of absence. The frozen evaluator uses an explicit failure taxonomy and terminal precedence; the publication table therefore warns that a lower-priority mechanism can be masked by a higher-priority terminal classification such as premature closure. Public benchmark questions and gold identifiers also create a contamination risk for future live search. A credential-free search-time self-exposure diagnostic over the frozen 400-task AutoResearchBench Wide candidate run is archived with the external results and reports a task rate of 0.0. That diagnostic does not claim absence of indirect, training-time, or live-provider contamination.

4.9. Decision rule

The paper-level rule is fail-closed. The offline companion can support a mechanism claim only at descriptive authority. External discovery superiority requires a denominator-valid external result that passes its archived matched-resource and safety rule. The TREC-COVID result fails that gate, the BEIR successors

remain negative mechanism diagnostics, and provider-invalid or unaccessed campaigns remain undetermined. None is promoted by analogy from the controlled world.

5. Results

5.1. Authority of the evidence

The evidence families answer different questions and are not pooled. The complete-gold index supplies descriptive mechanism diagnostics. The exact-contract battery tests agreement with authored closure labels. The TREC-COVID comparison supplies a descriptive external stress test and is adverse on recall and cost. The BEIR successors localize a stopping-rule failure without rescuing it. Provider-invalid or unaccessed campaigns remain undetermined.

5.2. Exact-contract closure test

The governed controller matched all 400 declared closure labels in the exact-contract battery. B1, the authored comparator composed of prior mechanisms, matched 250 labels, while B2 matched 100. The 150 B1 disagreements were exactly the unavailable, censored, and provider-invalid material-route cases. B1 passed every other archetype. The governed controller produced no false task closure and did not confuse the held-out unavailable non-material decoys with material obligations.

The governed-minus-B1 difference in exact scientific closure decision was $+0.375$. Its domain-stratified bootstrap 95% interval was $[0.3275, 0.4225]$, and the exact McNemar discordance counts were $b = 150$ and $c = 0$. These statistics describe the fixed authored battery; they are not population uncertainty or evidence of real-world retrieval performance.

B3, the information-equivalent comparator, also matched 400 of 400 labels and was decision-identical to the governed controller. This tie is load-bearing. It shows that the unresolved-route rule is portable to an enriched product and rules out an inherent-expressivity or centralized-ownership claim. The supported residual is narrower: when a declared material route is unavailable, censored, or provider-invalid, excluding it from the task-closure denominator violates the declared contract.

5.3. Descriptive TREC-COVID comparison is adverse

Across 50 paired TREC-COVID topics, the governed controller reaches mean recall@100 of 0.092642, compared with 0.110334 for the strongest comparator, BM25. The paired difference is -0.01769 , with a 95% percentile-bootstrap interval of $[-0.02729, -0.00906]$. Its lower bound extends below the post-hoc -0.02 non-inferiority reference margin. Mean reads rise from 85.52 to 235.8, an increase of 175.7%, rather than a 25% reduction. Recall and cost therefore both move against the proposed superiority interpretation.

The governed controller has higher nDCG@10, 0.566746 compared with 0.417992, for a paired difference of $+0.1488$. This favorable head-ranking result and the adverse recall and cost results describe a metric-dependent trade-off; the former does not establish overall superiority. Citation and restricted-provider routes are unavailable in the collection, so the controller also declines to declare task completeness on all 50 topics. The paper therefore makes no external retrieval-superiority claim.

5.4. BEIR successors retain the mechanism failure

The original BEIR route-aware stopping rule fails all three registered conditions. Its worst recall deficit from fusion is 0.101, its worst false-complete-rate excess over generic active stopping is 0.084, and it is cheaper than fusion at matched recall in only 5 of 15 corpus-depth cells. Favorable nDCG@10 values remain secondary and do not alter that terminal.

Density normalization leaves every ArguAna and SciFact decision identical to the original rule and changes NFCorpus without passing the full gate. All 15 cells still fail at least one condition. The subsequent query-conditional estimator again leaves all five ArguAna decisions and all five SciFact decisions unchanged. NFCorpus changes slightly but remains far from fusion at the deepest setting. The predeclared ArguAna falsifier therefore fires. Under the tested rule, both registered transformations are decision-inert on ArguAna. This is consistent with a rank-overlap marginal that supplies no usable decision signal there, but it does not establish that every alternative statistic would be uninformative. The no-rescue rule forbids a fourth estimator under the same study identity.

System	Recall	Premature closure	Pass	Undetermined	Fail
Governed system	0.979487	0.00	0.67	0.18	0.15
Protocol-driven review	0.666667	1.00	0.00	0.00	1.00
Sparse-dense hybrid	0.555556	1.00	0.00	0.00	1.00
One-pass retrieval augmented	0.555556	1.00	0.00	0.00	1.00
BM25 or keyword	0.333333	1.00	0.00	0.00	1.00
Dense-route comparator	0.333333	1.00	0.00	0.00	1.00
Agentic single route	0.333333	1.00	0.00	0.00	1.00

Table 1

Controlled-index descriptive comparison over 390 task variants. Failure for the governed system is honest budget exhaustion, not premature closure. Because variants are dependent within topic, no population or superiority inference is made.

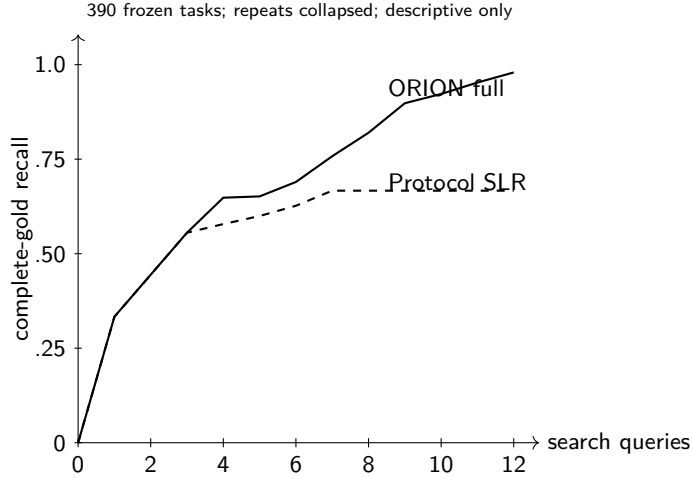


Figure 3: Cumulative complete-gold recall against host-recorded query count. Deterministic repeats are checked for identical traces and collapsed within task. Curves are descriptive means over 390 task variants.

5.5. Controlled-index comparison

The governed system attains mean complete-gold recall 0.979487 and precision 1.0. It passes 260 of 390 tasks, reaches honest budget exhaustion on 59, and leaves 71 undetermined because an unavailable route still holds unretrieved relevant material. It never certifies a censored task complete, so its premature-task-closure rate is zero. The strongest frozen comparator has mean recall 0.666667 and claims completion on all 390 tasks while reachable gold remains. The descriptive recall difference is therefore +0.31282, with a premature-closure difference of -1.0 . These quantities describe the authored controlled world rather than real-web performance.

The systems are identical through their first three search calls. The protocol-driven comparator reaches recall 0.666667 by query 7 and remains there through query 12. The governed system reaches 0.757835 by query 7 and 0.979487 by query 12. Figure 3 therefore localizes the descriptive gap to evidence acquired after the common prefix rather than to an initial advantage.

5.6. Mechanism ablations

The safety ablations fail in the intended directions. Removing the structural-distinctness check reduces mean recall to 0.444444 and produces premature closure on every task. Allowing route stop to certify task closure yields recall 0.333333 with the same 100% premature-closure rate. Giving a coverage diagnostic closure authority yields recall 0.555556 and again closes every task prematurely. These negative controls establish that the forbidden transitions are observable.

Removing content-identity deduplication raises mean duplicate processing from zero to 0.31453. It exhausts the read budget on 193 tasks, lowers pass rate from 0.666667 to 0.369231, and lowers mean recall to 0.939031. Removing the question coordinate leaves aggregate recall unchanged at 0.979487 but reduces

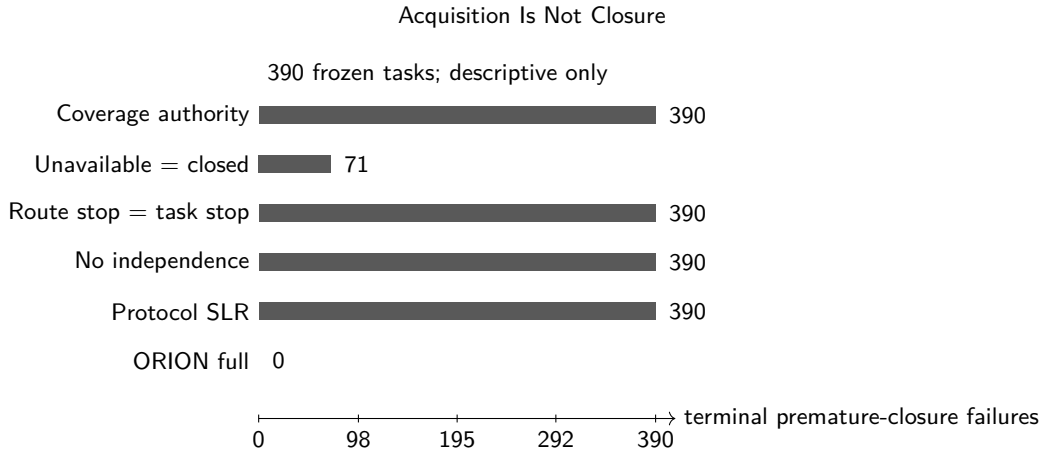


Figure 4: Terminal premature-closure classifications for selected safety controls. Bars are task counts in the fixed controlled suite; lower-precedence mechanisms can be masked by the evaluator’s terminal-failure ordering.

legitimate rereads on extraction-question-shift cases from 3.666667 to 1.0. Removing the unavailable-route open state also leaves aggregate recall unchanged, but turns 71 censored undetermined cases into false completion claims. These null-on-recall effects are retained because they isolate semantic safety rather than ranking gain.

The gold-defined route-stop replay records 13 false-positive route stops among 1,950 events. None becomes a task-level false closure because the material obligation remains open. Among 1,878 task-routes that reach oracle exhaustion, the evaluator observes zero route-stop false negatives. This null is reported without being generalized beyond the controlled world.

5.7. Bounded external probes and retained adverse results

The MetaSyn probe completes all 86 released test reviews under the official identifier-only evaluator. Macro retrieval recall is 0.7485 and final inclusion recall is 0.5651. Among 1,677 reference links, retrieval misses 598; screening then removes 343 of the 1,079 retrieved links, leaving 941 final false negatives. This localizes loss across acquisition and screening but does not exercise the complete controller.

A bounded AutoResearchBench Deep probe produces no official title matches on 600 tasks after a 9-of-9 judge control passes. A judge-free analysis likewise finds zero exact and substring recoveries among 10,850 candidate titles, with only eight half-token-overlap recoveries. The result is an acquisition-stage failure diagnostic, not a matched comparison. The SAGE setting remains unavailable as an executable arm because its audited release lacks the required runner, corpus, and licence combination.

The OpenAIRE and Crossref matched campaigns do not yield a valid null. One campaign records provider success of 0.666667 and another 0.715833, both below the frozen 0.90 validity gate. Their zero or near-zero paired differences are therefore transport-invalid and remain undetermined; they do not establish equivalence. A separate fresh 48-task campaign was not accessed and contributes no result.

Public screening diagnostics are also retained at their actual scope. In a five-review panel, the candidate trails the frozen active comparator on recall at 10% screened and work saved at 95% recall in all five reviews. A later 14-review public-development panel fails six locked gates, and a cross-fitted residual envelope harms both held-out reviews on which it activates. Subsequent source and identity audits stop before performance when their bindings fail or remain incomplete. These outcomes concern screening within already acquired pools; they neither support nor refute open-world acquisition superiority.

No external discovery-superiority claim is supported. The controlled-index and exact-contract results support a bounded authority mechanism; TREC-COVID and BEIR preserve adverse evidence; provider-invalid and unaccessed campaigns remain undetermined. Favorable secondary metrics never compensate for a failed primary gate.

Element	Ownership boundary
Acquisition and ranking	Lexical, reasoning-aware, field-aware, graph, and citation-expansion retrieval remain prior work; this paper does not claim a new general retriever.
Stopping and sufficiency	Utility, coverage, verification, conformal, and decision-sufficiency signals are drawn from cited prior work.
Memory and obligations	Question-conditioned memory, finite source certificates, and evidence-bound workflow obligations originate in cited prior work.
Residual tested here	A declared material route that is unavailable, censored, or provider-invalid remains in the task-closure denominator; the exact-contract battery isolates this aggregation rule.

Table 2

Contribution and prior-work boundary. The paper composes prior mechanisms and tests a narrower closure-authority residual.

6. Related work

Scientific-discovery benchmarks show that both targeted and open-ended literature search remain difficult for current agents [26, 11]. Acquisition methods are already heterogeneous. Strong lexical retrieval can outperform large-model retrievers inside deep-research workflows [10]; reasoning traces can become retrieval signals [3]; fielded admission and selective section fetching can improve the accuracy-context trade-off [24]; and bibliography expansion can recover studies missed by API-only search [21]. Crase fixes an inspectable citation-graph candidate set and stopping condition before inference, providing a directly adjacent bounded scholarly-search design [9]. These methods enlarge or bound the acquisition layer. They do not, without an additional argument, establish task completeness.

Automated evidence synthesis provides a second neighboring literature. MetaSyn separates retrieval from eligibility screening, making stage-specific false negatives visible [25]. AgentSLR evaluates an end-to-end systematic-review workflow over expert-annotated review tasks [16]. Technology-assisted review has long studied active screening protocols, autonomy, recall targets, and workload reduction [5, 6, 22]. The present work adopts stage attribution and fair-budget comparison, while focusing on the authority transition between a local acquisition outcome and a global closure claim.

Stopping research includes statistical stopping for systematic review [1, 13, 28], decision-theoretic continuation [8, 27], verification-aware completion [19, 4], conformal coverage for adaptive reasoning [30], and structured sufficiency or gap judgments [7, 14]. Finite obligation certificates and evidence-bound workflow transitions offer still stronger stopping mechanisms [2, 29]. We grant these mechanisms to the acquisition policy and ask a residual question: when a material route is unavailable, censored, or provider-invalid, may it disappear from the task-closure denominator? The exact-contract comparison is designed to change only that aggregation rule.

Capture-recapture methods and information foraging motivate, but do not settle, the same boundary. Capture-recapture estimates require defensible capture occasions and can be misleading under dependence [12, 20]. Information foraging describes the value of continuing or leaving a local information patch [17, 18]. In adaptive retrieval, however, a low-yield patch or a dependent route cannot be promoted into proof that the global literature has been exhausted. Question-conditioned memory likewise helps organize evidence [15], but the method here claims only the bookkeeping distinction between content identity and the question for which that content was processed.

7. Discussion and limitations

The controlled studies support a control-layer interpretation, not a retrieval leaderboard claim. In the complete-gold index, the governed controller maintains open obligations where the evaluator knows relevant material remains reachable. The ablations show that forbidden authority transitions are observable, while the exact-contract battery isolates nonconformance when unavailable material routes are excluded from the declared closure denominator. These results demonstrate implementation behavior under authored contracts. They do not estimate real-web difficulty or prove that the declared route registry contains every material

route. All gold labels, audits, and adjudications in this programme are same-programme and single-author work using AI assistance; no result reported here has been checked by an out-of-programme reviewer or an independent implementation outside this repository.

The offline companion is also underpowered against its frozen precision target, and the label follows from the plan rather than from any observed outcome: the generated suite facts record the plan's underpowered flag for this campaign as yes. The frozen statistical plan set a superiority margin of 0.03 for the primary standalone rate before outcomes were inspected, while the achieved half-width at the assembled variant count is 0.0496, so the margin is not met and no superiority decision is promoted from the offline companion. Repeats and variants do not change this: the honest statistical unit is the topic, and the suite's 390 task variants, 14 systems, and 3 deterministic repeats nest within 78 topics. The same preregistered arithmetic records the independent units a standalone rate requires at each precision level: 1068 at half-width 0.03, 385 at 0.05, 171 at 0.075, and 97 at 0.10. The assembled suite offers 390 task variants but only 78 independent topics, so the controlled-index numbers are read as mechanism diagnostics rather than as estimates with superiority resolution.

The TREC-COVID result is the strongest external stress test in the present paper because it compares the controller with public collection judgments [23]. It is descriptive rather than prospective: no separate pre-outcome record was found for the exact comparator-selection, margin, cost, and bootstrap rule. Recall is lower, its interval extends below the post-hoc -0.02 reference margin, and resource use moves in the wrong direction. The higher nDCG@10 indicates better top-ranked relevance on many topics, but it answers a different question. These opposing results do not establish external superiority.

Two separately frozen BEIR diagnostics sharpen the mechanism-level negative. Density normalization changes the nominal threshold but leaves every ArguAna decision identical to the original rule at all five depths. A subsequent query-conditional estimator again leaves all five ArguAna decisions unchanged, and SciFact is also unchanged; NFCorpus moves only slightly and remains well short of fusion. Under this decision rule, the two registered transformations are therefore inert on ArguAna. The observed values are consistent with a rank-overlap marginal that supplies no usable decision signal there, but the studies do not establish that every alternative statistic would be uninformative. The registered no-rescue rule forbids a fourth estimator under the same identity.

Several additional probes remain bounded or undetermined. A public MetaSyn probe quantifies retrieval and screening loss, but it does not exercise the complete controller. An AutoResearchBench Deep probe returns no official title matches after a successful judge control, which is a bounded negative diagnostic rather than a matched comparison. OpenAIRE/Crossref campaigns fail their provider-validity criterion; their zero paired differences cannot be read as equivalence. Public screening studies sometimes favor the frozen comparator over the candidate. The result-retention ledger keeps these outcomes visible without averaging incompatible authority levels.

The formal control rules also have limits. Structural route distinctness is conservative and provenance-based; it does not prove statistical independence. The closure decision is relative to declared material routes and reopen conditions. A previously unknown source family can therefore reopen a task. Content hashes support deduplication but do not decide scientific relevance. Finally, the controlled index supplies complete denominators by construction. Open-web completeness generally does not.

The OpenAIRE comparisons remain undetermined after provider-validity failure, and the separate fresh 48-task campaign was not accessed. Neither absence of an outcome nor a zero difference from invalid transport is treated as a null result. Broader live-provider evidence would support a successor claim, not the bounded filing claim evaluated here.

These limits shape the intended use. The control layer can sit beneath stronger retrievers, screening models, and stopping policies, because it constrains what their outputs mean rather than how they are produced. Its intended control function is to prevent provider failure, local flatness, high estimated coverage, or apparent sufficiency from becoming an unsupported global claim. Whether this conservatism improves real scientific decisions at acceptable cost requires a new prospectively frozen external study.

8. Data and code availability

The submission package contains the complete manuscript source, the controlled index generator and evaluator, the exact-contract case definitions and protected decision receipts, the 50 topic-level TREC-COVID result rows, and the frozen BEIR protocol and result records. It also includes machine-readable claim and result-retention ledgers that distinguish supported, adverse, null, and undetermined outcomes. A manifest binds each distributed file by SHA-256.

Preregistration

The study was preregistered and frozen before any outcome was accessed, under the identifier `P2.open-world-discovery.v1`. A reader checking whether a reported analysis was planned or chosen after the fact should compare against that freeze; it is the reason every number below can be traced to a decision made in advance of seeing it.

The controlled index and exact-contract battery are project-authored and can be replayed from the archived source. TREC-COVID and BEIR corpora are not redistributed because their original providers control access and licensing. The package instead supplies the pinned source identities, corpus digests where available, analysis code, and derived per-topic or per-query records needed to check the reported values after lawful retrieval from the original providers.

The journal reviewer package is anonymized and contains no author-identifying repository link. An attributed public archive locator and immutable release identifier will be supplied during filing or public release; these human- and portal-controlled identifiers are not synthesized in the manuscript. The OpenAIRE provider-invalid records and the unaccessed 48-task campaign remain in the result-retention ledger and are not represented as successful, null, or equivalent outcomes.

9. Reproducibility

Every quantitative statement in the filing manuscript is bound to a versioned result record. The controlled and exact-contract studies support same-source replay, not independent empirical replication. The original BEIR control was replayed on two machines before the density-normalized and query-conditional successors were interpreted; this is portability evidence for the same implementation, not a second implementation or a new sample. Third-party data access, provider availability, and future archive identifiers remain explicit prerequisites rather than being inferred from local checks.

10. Conclusion

Open-world scientific-literature discovery needs separate control of evidence acquisition and closure authority. The method developed here derives route identity from provenance, distinguishes content identity from question-conditioned processing, keeps route stopping local, and retains unresolved material routes as open obligations. Controlled and exact-contract tests show that these rules behave as designed. The descriptive TREC-COVID comparison shows lower recall and higher read cost; the favorable $nDCG@10$ difference does not establish overall superiority. The later density-normalized and query-conditional successors also leave every ArguAna stopping decision unchanged. Both registered transformations are decision-inert there under the tested rule. The present contribution is therefore a bounded, reproducible control method for representing acquisition authority. External retrieval benefit and open-world completeness remain unestablished.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI ChatGPT to support literature searching, source checking, reproducibility review, manuscript editing, and submission-package preparation. The tool was not treated as an author and did not replace scientific judgment. Result-bearing statements and citations were checked against the archived evidence and primary-source records. The authors remain responsible for the article's content.

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